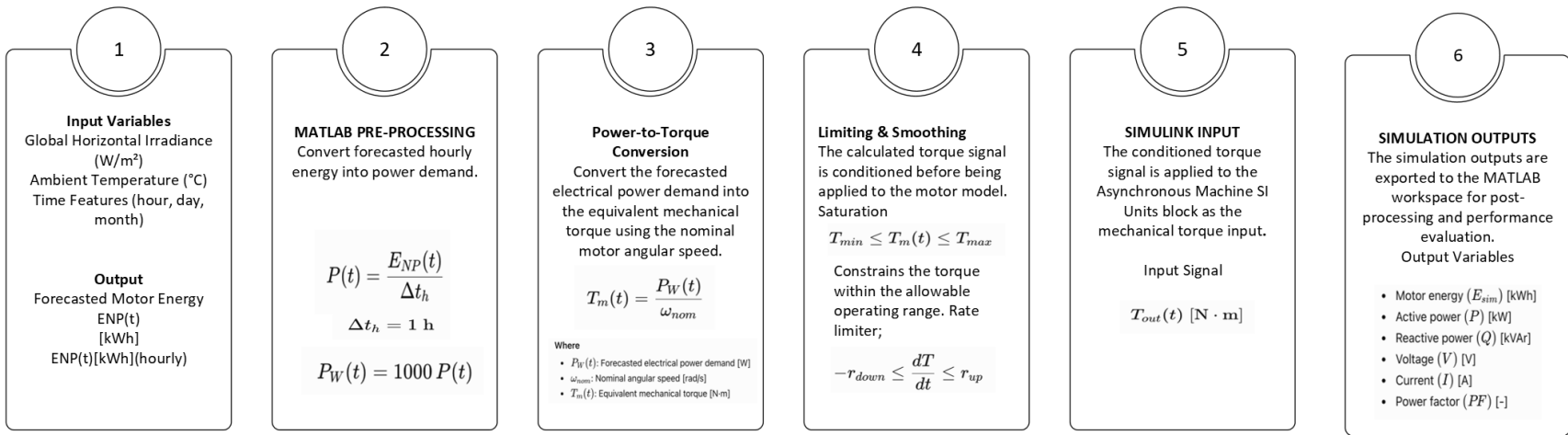
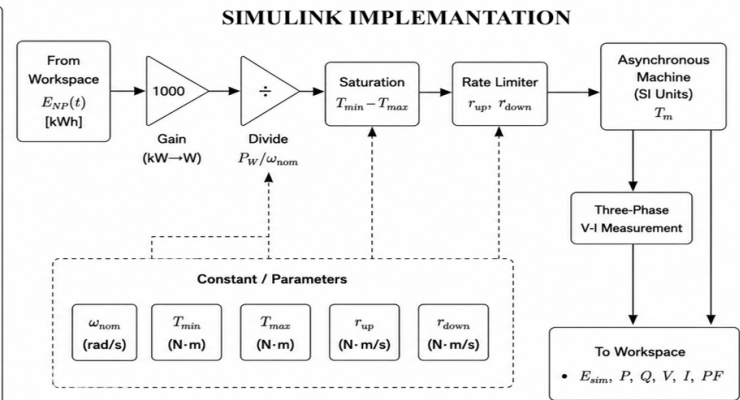
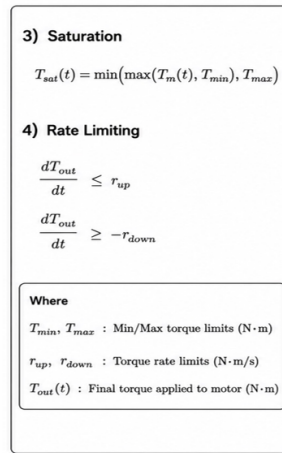
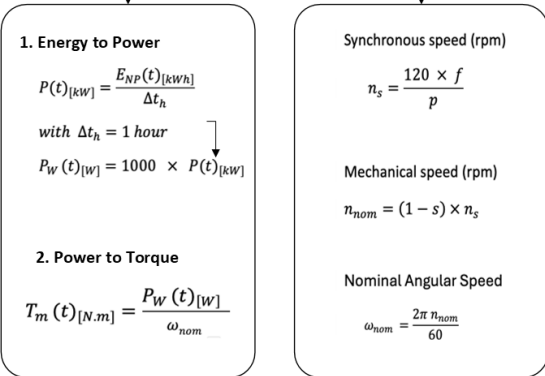




MATHEMATICAL FORMULATION



SIGNAL FLOW SUMMARY--KEY PARAMETER VALUES (EXAMPLE)

Signal Name	Description	Unit	Dimension	Source / Destination
$E_{NP}(t)$	Forecasted motor energy	kWh	N×1	NeuralProphet → MATLAB
$P(t)$	Power demand	kW	N×1	MATLAB
$P_W(t)$	Power demand	W	N×1	MATLAB
$T_m(t)$	Calculated torque	N·m	N×1	MATLAB
$T_{sat}(t)$	Saturated torque	N·m	N×1	Simulink
$T_{out}(t)$	Rate-limited torque	N·m	N×1	Simulink → Motor
$E_{sim}(t)$	Simulated motor energy	kWh	N×1	Simulink Output
$P(t)$	Active power	kW	N×1	Simulink Output
$Q(t)$	Reactive power	kVar	N×1	Simulink Output
$V(t)$	Voltage (RMS)	V	N×1	Simulink Output
$I(t)$	Current (RMS)	A	N×1	Simulink Output
$PF(t)$	Power factor	–	N×1	Simulink Output

N : Number of time steps (hourly)

Parameter	Symbol	Value	Unit	Notes
Nominal Power	P_n	22	kW	Nameplate
Line Voltage (RMS)	V_{LL}	400	V	3-phase
Frequency	f	50	Hz	Grid frequency
Poles	p	4	–	8-pole machine
Slip (rated)	s	0.02	p.u.	Typical
Nominal Speed	n_{nom}	1470	rpm	$(1-s) n_s$
Angular Speed	ω_{nom}	153.94	rad/s	$2\pi n_{nom}/60$
T_{min}	T_{min}	0	N·m	No negative torque
T_{max}	T_{max}	200*	N·m	Machine limit
r_{up}	r_{up}	500	N·m/s	Rate limit (up)
r_{down}	r_{down}	-500	N·m/s	Rate limit (down)

* Example value – adjust according to machine capability.

NOTES FOR REPRODUCIBILITY

- All calculations use hourly resolution ($\Delta t_h = 1 \text{ hour}$)
- Ensure unit consistency (kWh → kW → W → N·m).
- Adjust $T_{min}, T_{max}, r_{up}, r_{down}$ based on motor datasheet and dynamic response.
- This workflow is implemented in MATLAB R2025b / Simulink R2025b